MILITARY SPECIFICATION

AMPLIFIER, RADIO FREQUENCY, AM-7074/BRC

This specification is approved for use by the Space and Naval Warfare Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers manufacture and acceptance requirements for Radio Frequency (RF) Amplifier, AM-7074/BRC, hereinafter referred to as the RF amplifier.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standard(s). The following specifications and standard(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-B-636	Boxes, Shipping, Fiberboard
PPP-T-60	Tape, Packaging, Waterproof

MILITARY

MIL-E-16400	Electronic, Interior Communication And Navigation Equipment, Naval Ship And Shore: General Specification For

NAVAL UNDERWATER SYSTEMS CENTER (NUSC), NEW LONDON LABORATORY (NLL)

NUSC-C-342/4141-279	Cable, Radio Frequency, Buoyant, General Specification

STANDARD

MILITARY

MIL-STD-109	Quality Assurance Terms And Definitions

2.1.2 Other Government drawings (DWGs). The following other Government DWGs form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Space and Naval Warfare Systems Command (SPAWAR-8111), Washington, DC 20363-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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DWGs

NUSC

NUSC/NLL 02379005	Pin Contact
NUSC/NLL 02380006	Amplifier/Adapter Assembly Procedure
NUSC/NLL 02383001	Adapter, Connector (Pin Contact)
NUSC/NLL 02384001	Adapter, Connector (Socket Contact)

(Copies of specifications, standards, handbooks, DWGs, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

DWGs

SPEARS ASSOCIATES, INCORPORATED (SAI)

SAI C185050	Liner
SAI D185049	Housing Assembly
SAI E185033	Connector Plug, Electrical (Socket Contact) Cable Assembly R.F.
SAI E211024	Amplifier, R.F. AM 7074/BRC

(Copies of DWGs required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The RF amplifier shall be in accordance with MIL-E-16400 to the extent specified herein.

3.1.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.1.2 Description. The RF amplifier shall consist of the components specified in a through c:

a.	Housing Assembly:	SAI DWG D185049
b.	Liner:	SAI DWG C185050
c.	Amplifier, R.F. AM-7074/BRC:	SAI DWG E211024

3.2 Characteristics. RF amplifier characteristics shall be as specified in 3.2.1 through 3.2.3.5.

3.2.1 Performance. The RF amplifier shall interface with Antenna Element, AS-3435/BRC and RF Cable Assemblies, CG-3820/BRC, and CG-3819/BRC, and Antenna Transmission Line, AS-3437/BRC of Buoyant Cable Antenna Assembly, AS-3434(V)/BRC. The RF amplifier shall permit passive reception of RF signals over the very low frequency (VLF) and low frequency (LF) range and amplified reception of signals over the medium frequency (MF) and high frequency (HF) and very high frequency (VHF) and ultra high frequency (UHF) ranges. The RF amplifier shall be mechanically compatible with deployment and retrieval through Antenna Transfer Assemblies, AN/BRA-18( ) and AN/BRA-24( ).

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3.2.1.1 MF-UHF stability. The RF amplifier shall exhibit no internally generated parasitic oscillations. The same stability shall be exhibited when the amplifier is configured as part of a Buoyant Cable Antenna Assembly, AS-3434(V)5/BRC floating in a straight line in sea water.

3.2.1.2 MF-UHF voltage gain. The RF amplifier shall conform to the gain limits specified in TABLE I.

TABLE I. Gain requirements.

Frequency MHz 1/	Voltage gain dB 2/ ±3 dB
2	13
5	32
10	38
50	39
100	39
200	39
400	36

1/ Megahertz
2/ Decibels

3.2.1.3 MF-UHF loop gain. The loop gain at all frequencies between 0.3 MHz and 1250 MHz, where the amplifier produces positive gain, shall be no greater than -20 dB.

3.2.1.4 MF-UHF noise figure. The RF amplifier shall exhibit a noise figure no greater than 7 dB at all frequencies from 2 MHz to 400 MHz.

3.2.1.5 VLF and LF sensitivity. When tested in accordance with 4.5.8, the RF amplifier shall reduce the sensitivity by less than 2 dB at all frequencies from 10 kilohertz (kHz) to 100 kHz.

3.2.1.6 LF phase response. The phase response of the RF amplifier shall be such that a least mean square fit, straight line, drawn through the phase response plot over the frequency range 70 kHz to 130 kHz intercepts the phase shift axis within the interval 0 ±25 degrees. For the plot, the ordinate scale (phase shift) and the abscissa scale (frequency) shall both be linear. The deviation of the phase response from the least mean square fit line shall be less than 1 degree over the 70 kHz to 130 kHz frequency range and less than 3 degrees over the 50 kHz to 150 kHz frequency range.

3.2.1.7 VLF and LF amplitude response. The amplitude response of the RF amplifier shall conform to the requirements of TABLE II. (The maximum allowable loss at 10 kHz and at 100 kHz shall be 2 dB.)

TABLE II. VLF and amplitude response requirements.

Frequency range (kHz)	Amplitude variation (dB)
10 to 50	±3.0
50 to 70	±0.75
70 to 100	±0.4
(100)	0.0 (reference)
100 to 130	±4
130 to 150	±0.75
150 to 160	±1.0

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3.2.1.8 LF phase delay variation. The maximum and root-mean-square (rms) phase delay variations from the input to the output of the RF amplifier shall be no greater than 45 nanoseconds (ns) and 15 ns, respectively, in a 100-hour period over the frequency range 50 kHz to 160 kHz. Demonstration shall be by test in accordance with 4.5.11.

3.2.1.9 HF and UHF intermodulation products. The RF amplifier shall exhibit a level of intermodulation products at the output no greater than -30 decibels referred to one milliwatt (dBm).

3.2.1.10 HF and UHF gain compression. When tested in accordance with 4.5.5, the RF amplifier shall exhibit an output level of at least +8 dBm for a gain compression of 1 dB.

3.2.1.11 Power supply interruption resistance. The amplifier shall operate satisfactorily after being subjected to the direct current (DC) power on-off and short circuit cycles of 4.5.18.

3.2.2 Physical characteristics. RF amplifier physical characteristics shall be as specified in 3.2.2.1 through 3.2.2.2.

3.2.2.1 Weight and dimensions. The amplifier section shall fit the housing assembly shown in SAI DWG D185049 and shall provide the connector interfaces shown in SAI DWG E211024. The maximum length of the housing assembly shall be 8.25 inches (in.). The liner specified in SAI DWG C185050 is an integral part of the RF amplifier. The weight of the amplifier section including chassis and connector shall not exceed 110 grams.

3.2.2.1.1 Diameter. The diameter of the amplifier housing shall be 0.650 in. ±0.02 in.

3.2.2.2 Health and safety. Health and safety considerations require avoidance of toxicological substance such as encapsulants, adhesives, or preservatives which could contaminate the closed environment of a submarine. The materials specified in SAI DWGs D185049 and E211024 shall be acceptable. Any substitutes shall be approved by the Government.

3.2.3 Environmental conditions. Environmental conditions shall be as specified in 3.2.3.1 through 3.2.3.5.

3.2.3.1 Temperature. The amplifier shall operate over the temperature range of -10° Celsius (C) to +50°C and shall withstand shipment and storage over the temperature range -40°C to +70°C.

3.2.3.2 Pressure resistance. When tested in accordance with 4.5.14, the RF amplifier assembly shall show no sign of damage and show no evidence of flooding and exhibit no electrical degradation.

3.2.3.3 Flexibility. When tested in accordance with 4.5.1, the RF amplifier shall exhibit no physical damage as revealed by surface examination and pressure resistance tests, and no electrical degradation.

3.2.3.4 Durability. When tested in accordance with 4.5.15, the RF amplifier shall exhibit no physical damage and no evidence of flooding as revealed by surface examination and no electrical degradation.

3.2.3.5 Tensile strength. When tested in accordance with 4.5.16, the RF amplifier assembled with suitable test cables shall withstand a static tensile load along its axis with no evidence of cracking or other damage. The elongation of the housing shall be less than 1.6 percent.

3.3 Design and construction. SAI DWG E211024 shall be used as a guide to achieving the functional performance specified herein. The RF amplifier section shall be assembled in a housing fabricated in accordance with SAI DWG D185049.

3.3.1 Electrical design. The polarity and grounding of circuits shall be as specified to ensure compatibility with couplers and to minimize electrolysis of exposed antenna components. The RF amplifier shall be designed to operate from a constant voltage source of 27 volts direct current (VDC) connected with the positive side to the center conductor and the negative side to the outer conductor of RF Cable Assembly, CG-3819/BRC, with which it interfaces. The current drawn by the RF amplifier shall be less than 100 milliamperes (mA) ±10 mA. The output of the RF amplifier shall be connected between the center conductor and the outer conductor of the RF cable assembly with which it interfaces. The outer conductor on the output side of the RF amplifier is grounded to sea water at the electrical plug connector (socket contact) as specified in SAI DWG E185033. The amplifier input shall have two diodes, one of each polarity, shunted to ground for electromagnetic pulse protection.

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3.3.2 Materials and parts. Materials and parts shall be in accordance with MIL-E-16400. Materials and parts listed on the DWGs are approved for the application shown. The DWGs and specifications for the connectors and housing of the RF amplifier shall be followed for form and fit in order to assure interchangeability.

3.4 Workmanship. The RF amplifier shall be manufactured with care using the materials, dimensions, and surface finishes specified. Correct arrangement, hookup, and fastening of all components during fabrication shall be assured. The molding procedure for the housing is particularly critical and shall be performed with careful attention to spacing, lay, and tension of reinforcing members. The completed RF amplifier shall be clean and free of sand, grit, burrs, cuts, and gouges.

3.4.1 Workmanship screen. The RF amplifier shall withstand a defect detection vibration screen of random type vibration at $0.04g^2/Hz$ ±3 dB from 80 Hz to 350 Hz, and temperature cycling as specified in 4.6.1.2.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall conform to all requirements of Section 3 and Section 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of, but is not limited to, a) surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied, b) Government product inspection to measure quality of the product to be offered for acceptance, and c) Government inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been taken or until conformance of the product to prescribed criteria has been demonstrated.

4.1.3 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

4.2 Classification of inspection. The inspection requirements specified herein are classified as specified in a through c:

a. First article inspection (see 4.3)
b. Quality assurance inspection:
   1. Production inspection (Group A) (see 4.4.1)
   2. Production control inspection (Group B) (see 4.4.2)
c. Inspection of preparation for delivery (see 4.7)

4.3 First article inspection. Unless otherwise specified (see 6.2.1), one unit shall be required for first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the tests specified in TABLE III.

4.4 Quality conformance inspection. Quality conformance inspection shall be as specified in 4.4.1 through 4.4.2.2.

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4.4.1 Production inspection (Group A). Production inspection shall be conducted on each RF amplifier offered for delivery. Production inspection shall comprise such examinations and tests which will prove the workmanship and reveal omissions and errors of the production process such as functional and performance tests at a limited number of points in the required range, tests which detect deviations from design, and tests which detect hidden defects of materials. Production inspection shall include the examinations and tests specified in TABLE III, Group A.

4.4.1.1 Nonconforming production units. If more than three units per 50 units tested fail the production inspection specified in 4.4.1, the contractor shall immediately investigate the cause of failure to determine if the redesign permitted in 3.2.2.1 has affected reliability (see 6.2.1c).

TABLE III. Examination and tests.

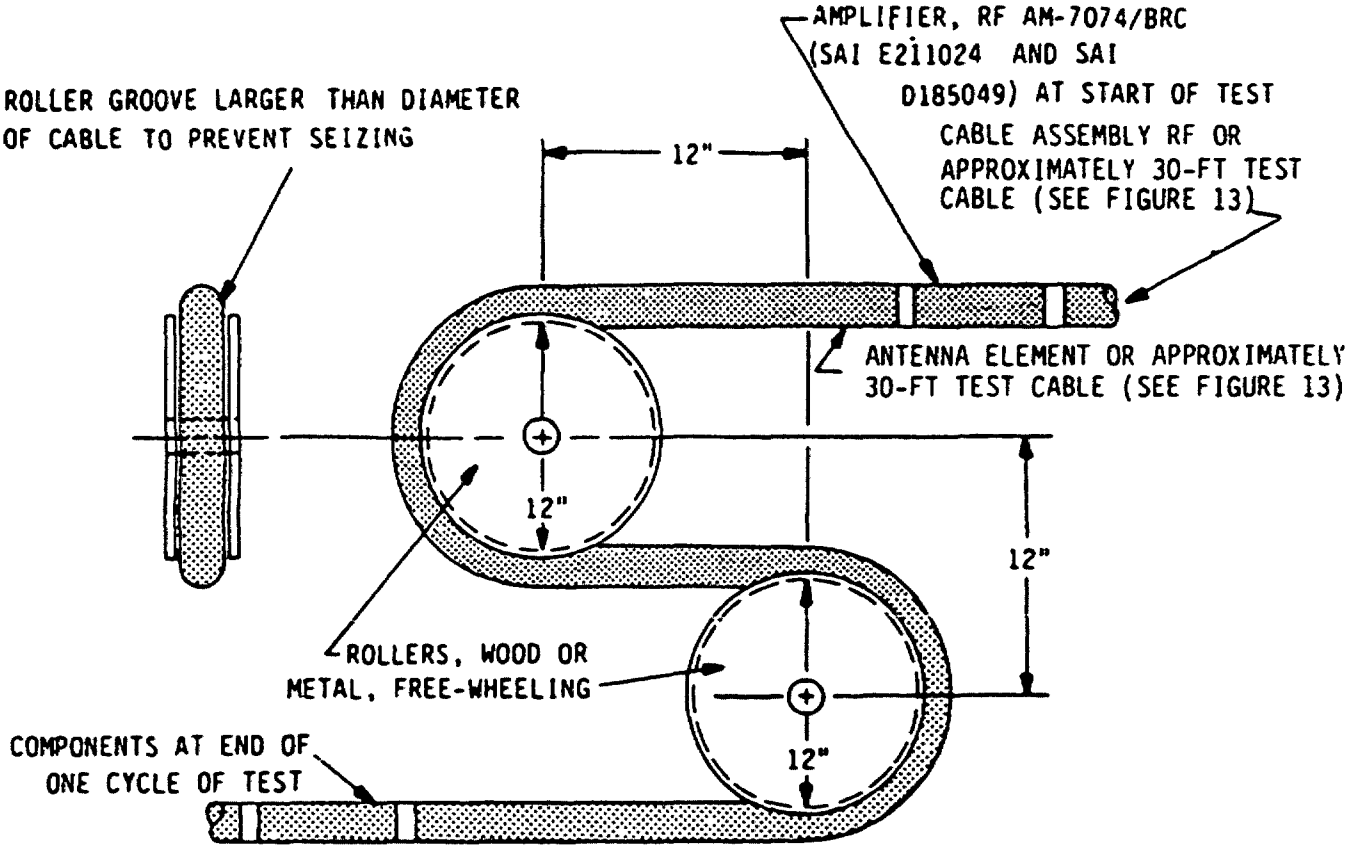
Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection	
				Group A	Group B
Workmanship screen	3.4.1	4.6.1	X	X	
Flexibility	3.2.3.3	4.5.1	X	X	
MF-UHF stability	3.2.1.1	4.5.4	X	X	
MF-UHF voltage gain	3.2.1.2	4.5.2	X	X 1/	
MF-UHF loop gain	3.2.1.3	4.5.3	X		X
HF and UHF gain compression	3.2.1.10	4.5.5	X		
MF-UHF noise figure	3.2.1.4	4.5.7	X		X
VLF and LF sensitivity	3.2.1.5	4.5.8	X	X	
HF and UHF intermodulation	3.2.1.9	4.5.6	X		
Pressure resistance	3.2.3.2	4.5.14	X	X	
Weight and dimensions	3.2.2.1	4.5.13	X		
Design and construction	3.3	4.5.13	X		
Materials and parts	3.3.2	4.5.13	X		
Durability	3.2.3.4	4.5.15	X		
Tensile strength	3.2.3.5	4.5.16	X		X
Diameter	3.2.2.1.1	4.5.17	X		X
Temperature	3.2.3.1	4.5	X		
Workmanship	3.4	4.6	X	X	
LF phase response	3.2.1.6	4.5.10	X		
VLF and LF amplitude response	3.2.1.7	4.5.9	X	X 2/	
LF phase delay variation	3.2.1.8	4.5.11	X		X
Power supply interruption resistance	3.2.1.11	4.5.18	X		

1/ 2 MHz, 10 MHz, 50 MHz, and 200 MHz only
2/ 10 kHz, 50 kHz, 100 kHz, and 150 kHz only

4.4.2 Production control inspection (Group B). Production control inspection shall be conducted on a sampling basis as specified in 4.4.2.1. Production control inspection shall consist of examinations and tests which encompass functional and performance tests throughout the entire range of operations and tests to determine deviations in the processes and equipment required to fabricate the RF amplifier. These tests shall be performed on the complete RF amplifier as offered for delivery. Production control inspection shall include the examinations and tests specified in TABLE III, Group B.

4.4.2.1 Sampling plan for production control inspection. RF amplifiers submitted for production control sampling tests shall be selected by the Government Quality Assurance Representative (QAR) and shall be representative of current production. One RF amplifier shall be selected from each successive 25 production RF amplifiers or fraction thereof.

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NOTE: Cables and connectors can be completed antenna or test sections sufficient to allow amplifier to be drawn around rollers. Use sufficient tension to bend amplifier into close contact with rollers. Ensure contact through at least 180 degrees on each roller.

FIGURE 1. Flexibility test setup.

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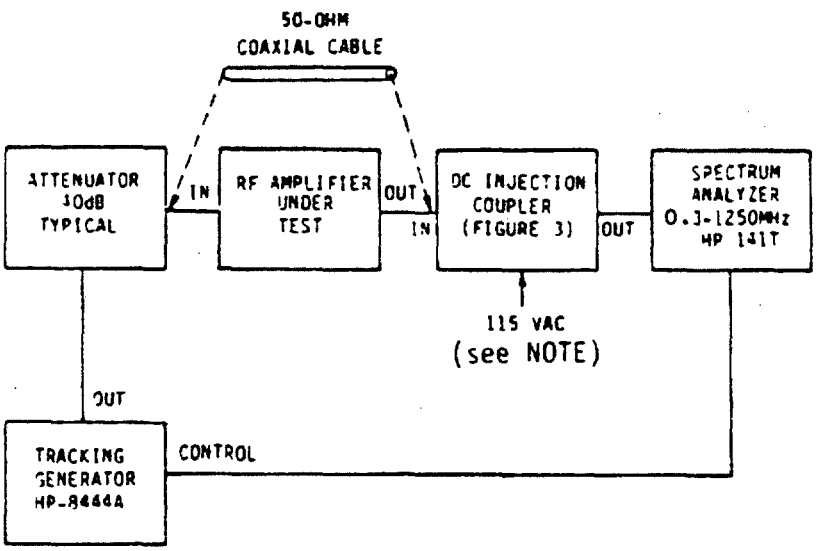
4.4.2.2 Nonconformance corrective action and retest. If a sample unit fails the inspection specified in 4.4.2, the contractor shall immediately investigate the cause of failure and shall report to the QAR the results thereof and details of the corrective action taken to correct units of product which were manufactured under the same conditions, with the same materials, processes, and so forth. No samples shall be delivered as production items under the same contract.

4.5 Test methods. To the extent specified herein, the examinations and tests specified in TABLE V should be performed in accordance with the test methods specified in MIL-E-16400. Testing shall be performed at ambient pressure and room temperature unless otherwise specified herein.

4.5.1 Flexibility test. The RF amplifier, assembled with test cables, shall be flexed through an S-bend path device as shown in FIGURE 1, three times in each direction, at 60 feet (ft) per minute. The RF amplifier shall subsequently pass all the other tests and inspections specified herein.

4.5.2 MF-UHF voltage gain tests. The RF amplifier assembled with test adapters, NUSC/NLL DWGs 02383001 and 02384001, at each end, shall be tested for voltage gain as specified in a through g:

a. Connect the RF amplifier with test adapters in the test setup shown in FIGURE 2.
b. Remove DC power from the DC injection coupler and then remove the RF amplifier from the test setup, replacing the RF amplifier with a 50-ohm cable.
c. Set the step attenuator to 40 dB and the tracking generator output to minimum (-10 dBm) to avoid saturation of the amplifier.
d. Select the frequency range to be displayed and set the spectrum analyzer for a convenient reference level near the bottom of the screen.
e. Replace the 50-ohm cable with the RF amplifier, apply DC power, and use a camera or plotter to observe the gain versus frequency curve.
f. Repeat b, c, d, and e for appropriate frequency scans until the gain over the entire range from 0.3 MHz to 1250 MHz has been observed.
g. Compare the test results with the gain requirements specified in TABLE 1 to verify conformance to 3.2.1.2. Observe the frequency (near 1 MHz) where the gain is exactly 0 dB. This frequency and the gain data above 400 MHz are needed for the loop gain test of 4.5.3.



NOTE: Volts alternating current

FIGURE 2. MF-UHF voltage gain test setup.

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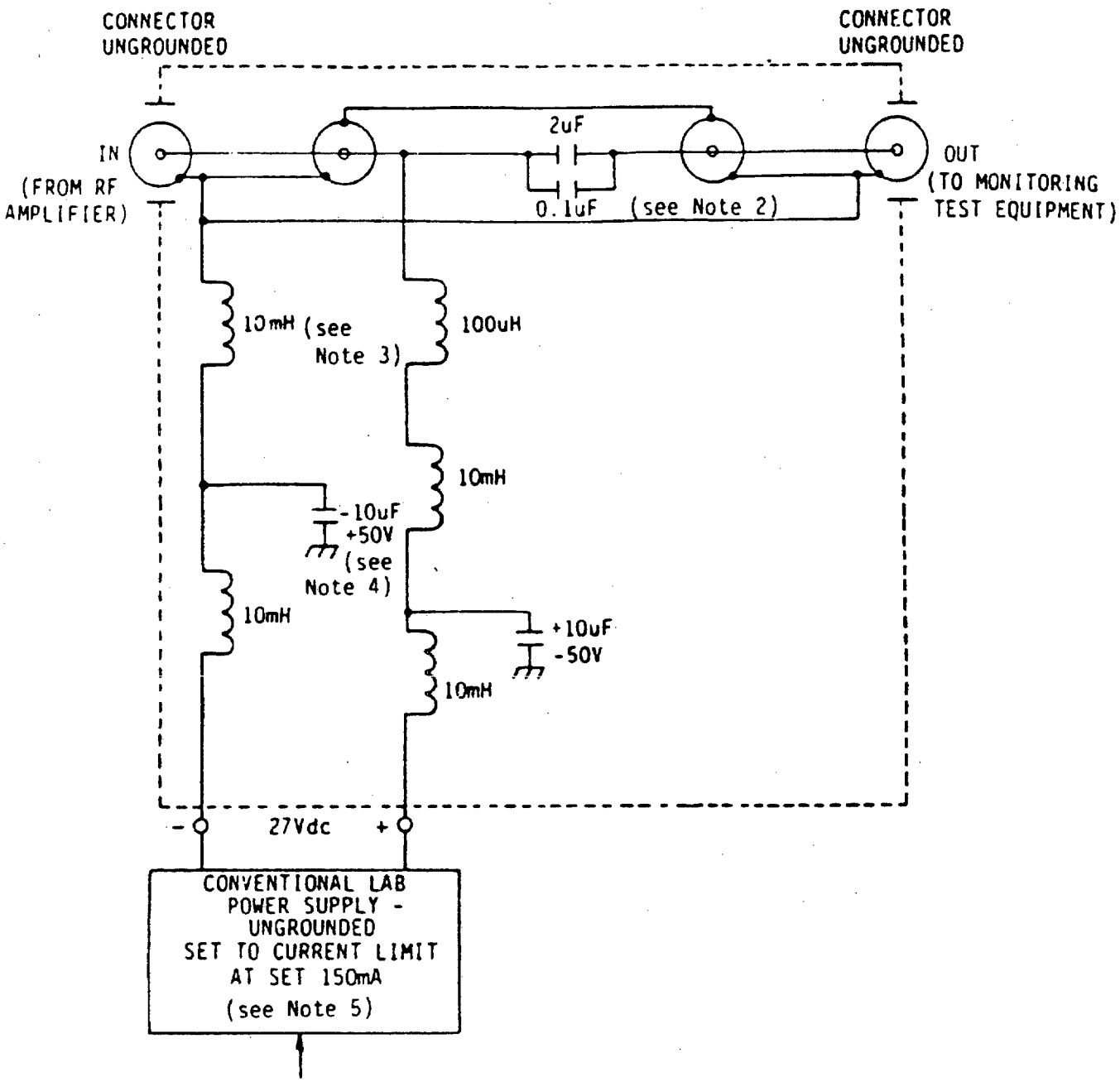
4.5.3 MF-UHF loop gain test. The loop gain shall be determined by measuring the loss through the VLF and LF bypass network in an unencapsulated RF amplifier section, modified as specified in a and in accordance with the procedure specified in b through i to verify conformance to 3.2.1.3:

a. Disconnect one RF lead at the input or output of stage two of the MF-UHF amplifier stages to eliminate any amplifier feedthrough.
b. Assemble the modified amplifier section in its housing together with test adapters and place in the test setup of FIGURE 2 and FIGURE 3, omitting the DC injection coupler and the attenuator.
c. Set the attenuator to 10 dB and the tracking generator to -10 dBm.
d. Replace the modified RF amplifier with a 50-ohm cable.
e. Select the frequency range to be displayed and set the spectrum analyzer for a convenient reference level near the top of the screen. The noise level with the tracking generator OFF shall be at least 50 dB below the reference.
f. Replace the 50-ohm cable with the modified RF amplifier and observe the loss (negative gain) versus frequency curve in the same manner as specified in 4.5.2e.
g. Repeat d, e, and f for appropriate frequency scans until the gain over the entire range of 0.3 MHz to 1250 MHz has been observed.
h. Add the data obtained in f and g to the data obtained in the test of 4.5.2 to obtain the loop gain.
i. Compare the test results with the requirement of 3.2.1.3. Note that greater than -20 dB means increasing toward positive values such as -10 dB, -5 dB, and so forth.

4.5.4 MF-UHF stability test. The RF amplifier, assembled with test adapters at each end, shall be tested for stability as specified in a through d to verify conformance to 3.2.1.1:

a. Connect the RF amplifier in the test setup of FIGURE 3 and FIGURE 4 with DC power applied to the DC injection coupler.
b. Scan over the 0.3-MHz to 1250-MHz frequency range to detect if signals are present. No signals should be observed.
c. Repeat step b with the 50-ohm load removed from the input of the RF amplifier.
d. For first article only, repeat b with the RF amplifier assembled as part of a Buoyant Cable Antenna Assembly, AS-3434(V)5/BRC floating in sea water in a straight line. The AS-3434(V)5/BRC consists of an Antenna Element, AS-3435/BRC, RF Amplifier, AM-7074/BRC, and RF Cable Assembly, CG-3819/BRC.

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NOTES:
1. This coupler should exhibit a loss of less than 1 dB when inserted in a 50Ω line between a signal source and a receiver, with coupler power off, over the frequency range 2 MHz to 400 MHz.
2. Microfarad.
3. Millihenries.
4. Volts.
5. Milliamperes.

FIGURE 3. DC injection coupler.

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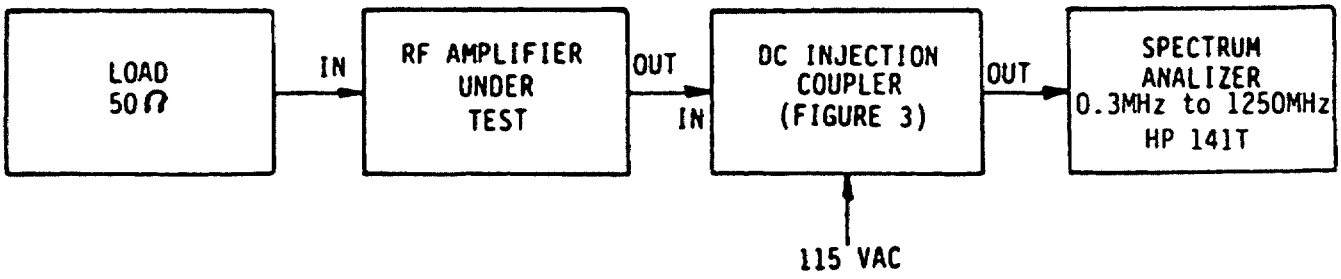


FIGURE 4. MF-UHF stability test setup.

4.5.5 HF and UHF gain compression test. The RF amplifier, assembled with test adapters at each end, shall be tested for dynamic range as specified in a through e to verify conformance to 3.2.1.10:

a. Connect the RF amplifier in the test setup of FIGURE 3 and FIGURE 5 with DC power applied to the DC injection coupler and with the signal generator and spectrum analyzer set at 10 MHz.

b. Adjust the signal generator output to produce 40-dBm in the spectrum analyzer display with the external step attenuators set at 0 dB.

c. Increase the signal generator output by 10 dB and switch in 10 dB of loss with the step attenuators. Continue increasing the signal generator output level while adding an equal amount of attenuation with the attenuators, in 10 dB and then 1 dB steps until the level in the spectrum analyzer display has dropped to -41 dBm.

d. Compare the test results with the requirements of 3.2.1.10: The output level of the RF amplifier for 1 dB of gain compression is determined as follows:

Output level (dBm) = -41 plus the setting of 1 dB step attenuator plus the setting of 10 dB step attenuator.

e. Repeat b to d with the signal generator and spectrum analyzer set at 400 MHz.

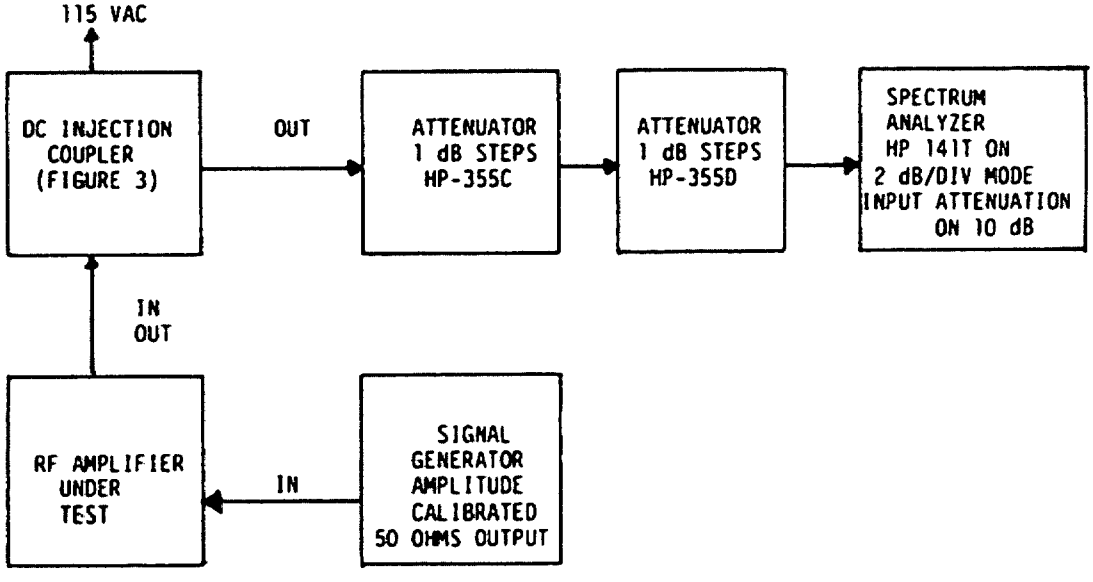


FIGURE 5. HF and UHF gain compression test setup.

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4.5.6 HF and UHF intermodulation test. The RF amplifier, assembled with test adapter at each end, shall be tested for intermodulation products as specified in a through e to verify conformance to 3.2.1.9:

a. Connect the amplifier in the test setup of FIGURE 3 and FIGURE 6 with DC power applied to the DC injection coupler, the spectrum analyzer centered at 10 MHz, signal generator f_1 set to 9.5 MHz, and signal generator f_2 set to 10.5 MHz.

b. Adjust each signal generator amplitude to produce a fundamental frequency signal level of 0 dBm on the analyzer screen.

c. Observe the levels of the third order intermodulation products, that is, signal levels on the analyzer screen at frequencies ($2f_1 - f_2$ and $2f_2 - f_1$).

d. Repeat b and c with the spectrum analyzer centered at 400 MHz, signal generator f_1 set to 399 MHz, and signal generator f_2 set to 401 MHz.

e. Compare the test results with the requirements of 3.2.1.9.

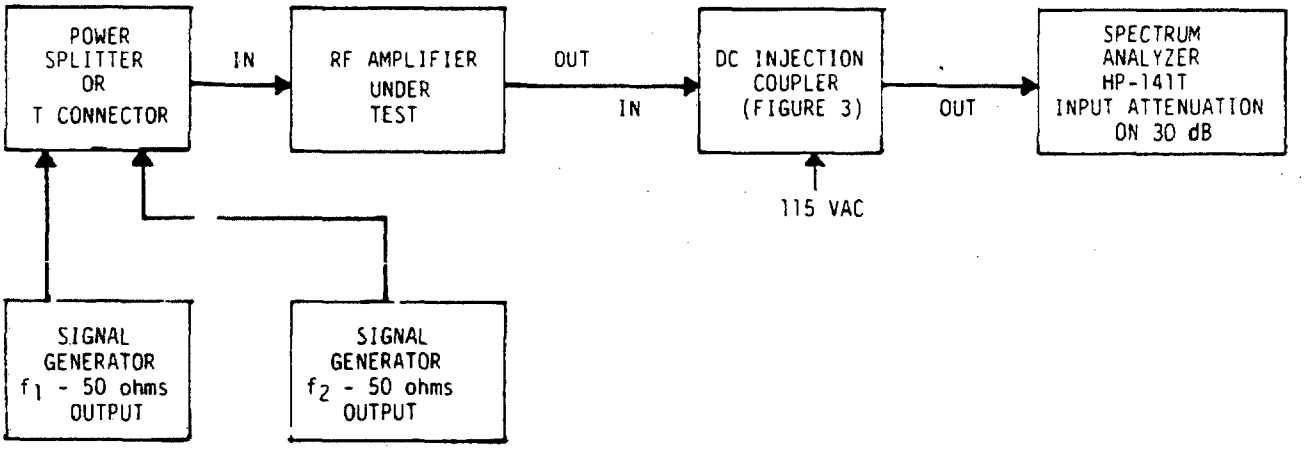


FIGURE 6. HF and UHF intermodulation test setup.

4.5.7 MF-UHF noise figure test. The RF amplifier, assembled with test adapters at each end, shall be tested for noise figure as specified in a through e to verify conformance to 3.2.1.4:

a. Connect the RF amplifier in the test setup as shown in FIGURE 3 and FIGURE 7. A test setup using an automatic noise figure meter may be substituted if available. Apply DC power to the DC injection coupler.

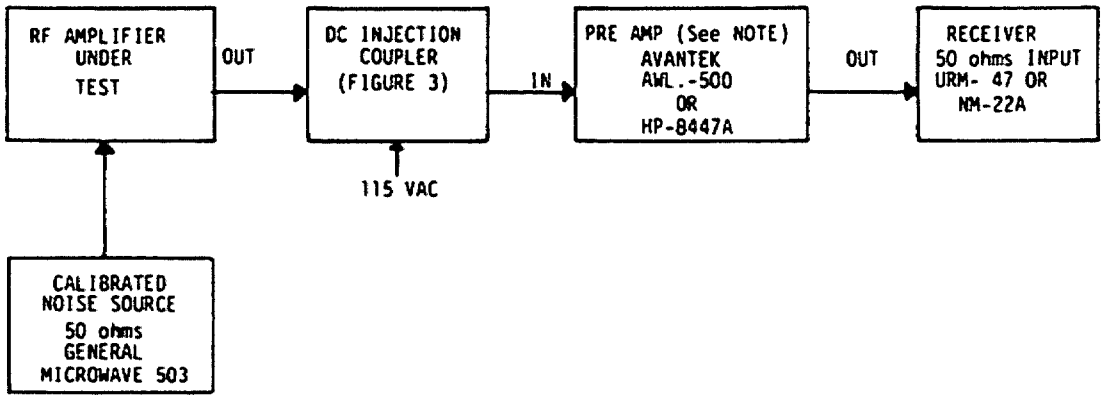
b. Turn the noise source to minimum and set the receiver to a convenient reference level at 3 MHz.

c. Increase the output of the noise source until the receiver noise is 3 dB above the reference level and then observe the noise figure indicated on the noise source.

d. Repeat b and c at 5 MHz, 10 MHz, 50 MHz, 200 MHz, and 400 MHz.

e. Compare the results with the requirement of 3.2.1.4.

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NOTE: Use preamplifier for 2-MHz measurement only.

FIGURE 7. MF-UHF noise figure test setup.

4.5.8 VLF and LF sensitivity test. The RF amplifier, assembled with test adapters at each end, shall be tested for sensitivity as specified in a through g to verify conformance to 3.2.1.5.

a. Connect the RF amplifier into the test setup of FIGURE 3, FIGURE 8, FIGURE 9, and FIGURE 10 without the DC injection coupler.

b. Remove the RF amplifier and connect the grounded dummy antenna directly to the DC injection coupler.

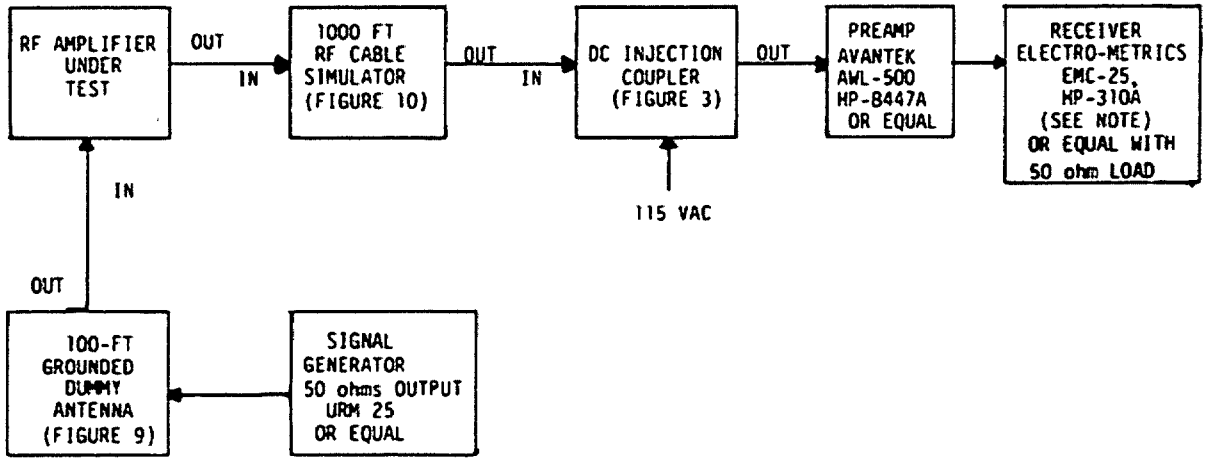
c. Disconnect the signal generator and set a convenient noise reference level on the receiver at 20 kHz.

d. Connect the signal generator and adjust the output to produce a signal level 20 dB above the reference level. Observe the output level of the signal generator. In this case, a level no greater than 6 microvolts indicates that the test setup is working properly.

e. Repeat c and d at 10 kHz, 30 kHz, and 100 kHz.

f. Connect the RF amplifier into the circuit and repeat c, d, and e both with DC power applied to and removed from the DC injection coupler.

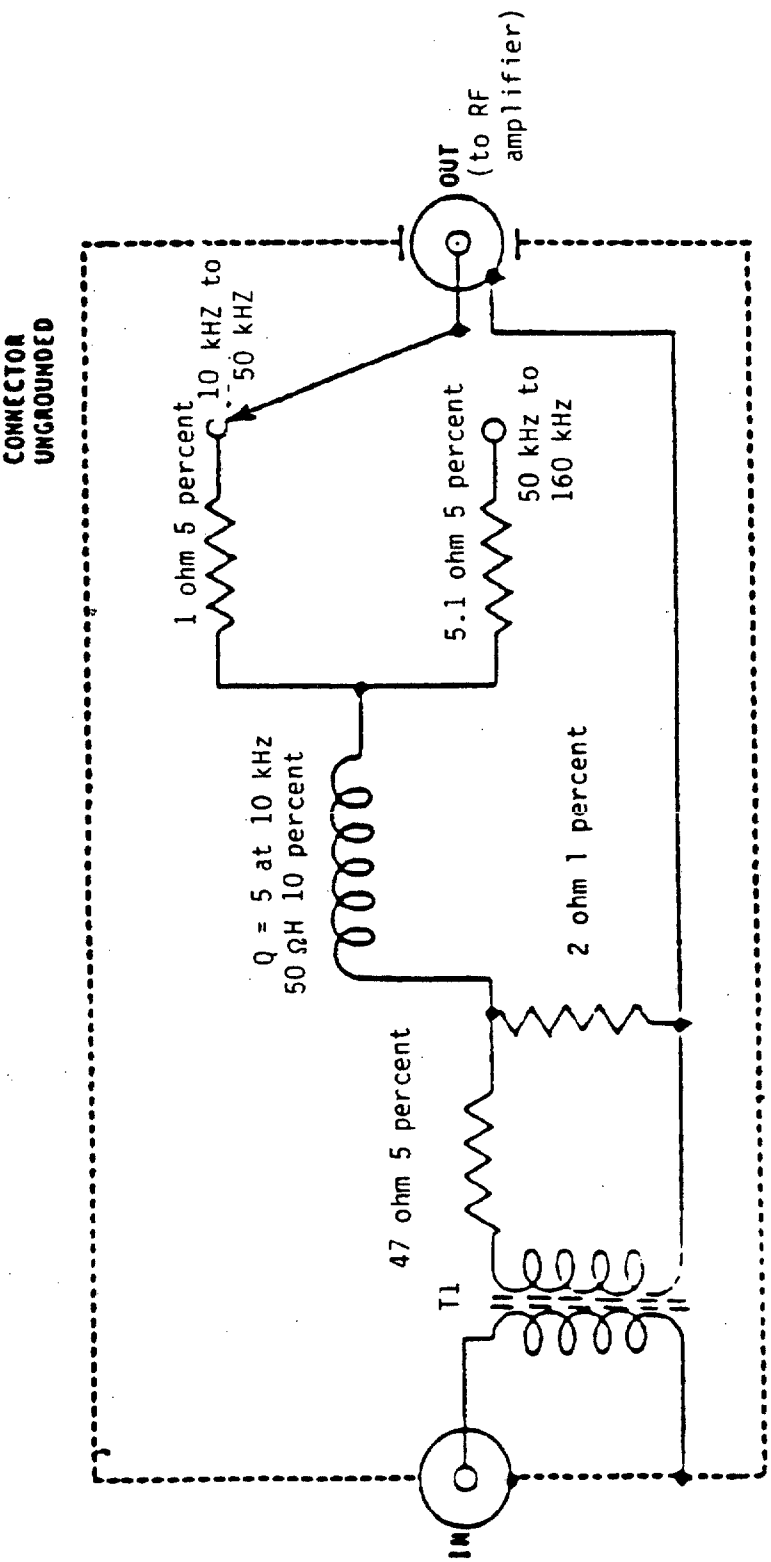
g. Divide each voltage obtained in f by the voltage obtained at the same frequency in d and e, and convert the result to dB. Compare results with the requirements of 3.2.1.5.



NOTE: The HP-310A may cause high frequency instability (which can generate VLF noise) in the power on mode.

FIGURE 8. VLF and LF sensitivity test setup.

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T1 = 50 ohm: 50 ohm TRANSFORMER WITH D-C ISOLATION

VARI-L MODEL LF-412 Z-MATCH OR EQUAL

FIGURE 9. 100-ft grounded dummy antenna.

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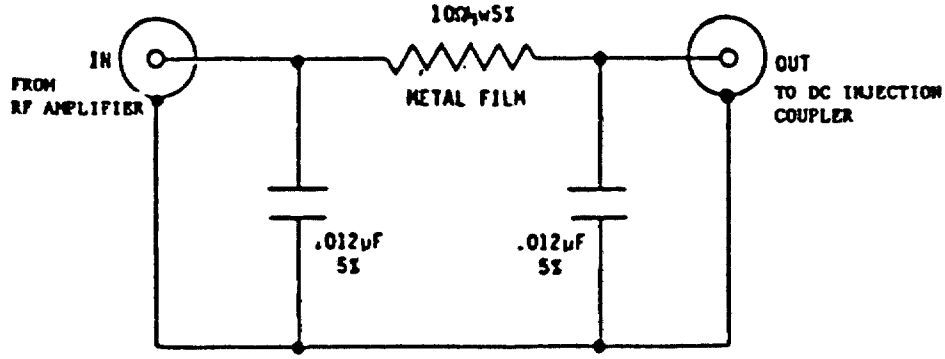


FIGURE 10. VLF and LF 1000-ft RF cable, (CG-3819/BRC).

4.5.9 VLF and LF amplitude response. The RF amplifier, assembled with test adapters at each end, shall be tested for amplitude response as specified in a through g to verify conformance to 3.2.1.7:

a. Connect the RF amplifier in the test setup of FIGURE 3, FIGURE 9, and FIGURE 11 without DC power supplied to the DC injection coupler.
b. Replace the RF amplifier by a 50-ohm cable.
c. Set the receiver and signal generator to 10 kHz and set a reference level on the wave analyzer.
d. Substitute the RF amplifier for the 50-ohm cable and read the new level on the wave analyzer.
e. Observe the difference between the level obtained in d and the reference level. This difference in level is the gain (or loss).
f. Repeat a through e at 10-kHz intervals up to 160 kHz both without and with DC power applied to the DC injection coupler.
g. Plot two curves showing loss versus frequency on linear graph paper for the two conditions, without and with DC power applied. Compare flatness with the requirements of 3.2.1.7.

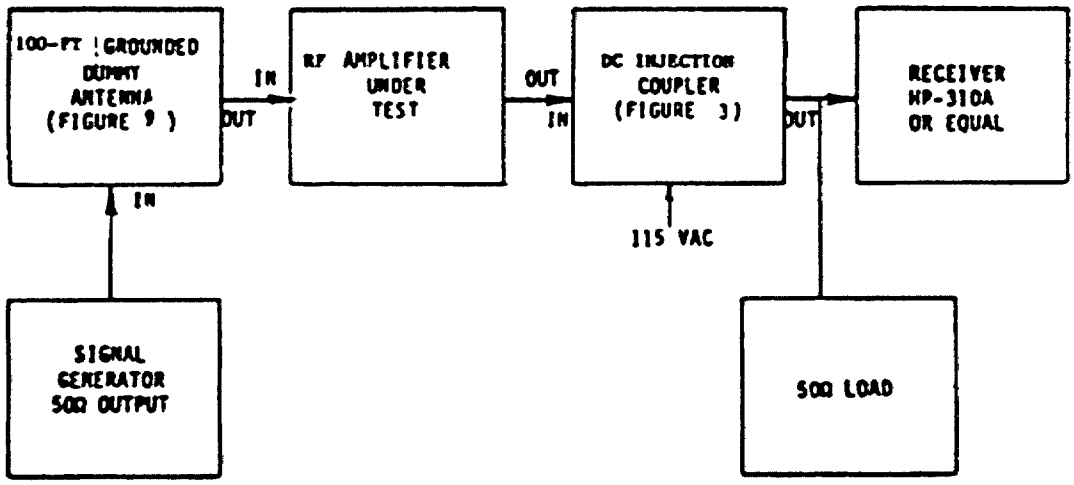
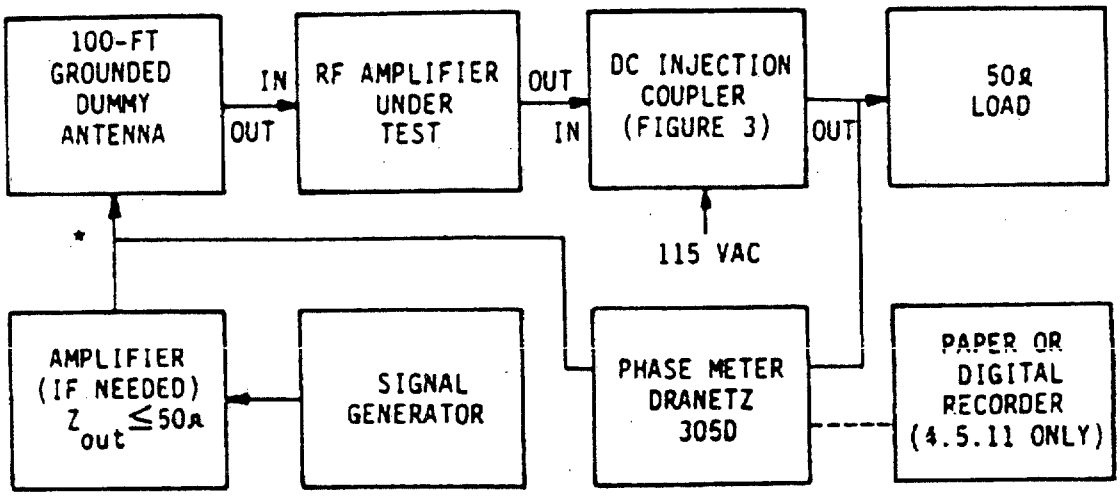


FIGURE 11. VLF and LF amplitude response test setup.

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4.5.10 LF phase response. The RF amplifier, assembled with test adapters at each end, shall be tested for phase response as specified in a through f to verify conformance to 3.2.1.6:

a. Connect the RF amplifier in the test setup of FIGURE 3, FIGURE 9, and FIGURE 12 without DC power applied to the DC injection coupler. Remove the transformer from the dummy antenna for this test (see FIGURE 3).
b. Adjust the signal generator to a level suitable for both inputs of the phasemeter.
c. Starting at 50 kHz, record phase shift in 10-kHz increments up to 150 kHz.
d. Plot phase shift versus frequency on linear graph paper. Add the least mean square fit straight line for the range 70 kHz to 130 kHz that is determined using the data observed in c.
e. Repeat b, c, and d with DC power applied to the DC injection coupler. Allow a 15-minute warmup period for the RF amplifier.
f. Using the data of d or e, determine the deviation of measured phase shift, from the least mean square fit line, for the 70-kHz to 130-kHz range and the 50-kHz to 150-kHz range. Compare the results with the requirement of 3.2.1.6.



NOTE: Hold signal amplitude constant at this point.

FIGURE 12. LF phase response and phase delay variation test setup.

4.5.11 LF phase delay variation. The RF amplifier, assembled with test adapters at each end, shall be tested for phase delay variation as specified in a through e to verify conformance to 3.2.1.8:

a. With the test setup of FIGURE 12 in an environmental chamber and without DC power applied to the DC injection coupler, perform a, b, and c of the phase response test specified in 4.5.10 at room temperature and then at 0°C.
b. With the temperature at 0°C and the frequency at 100 kHz, set the chamber temperature to increase at a rate of 1°C per hour. Observe phase data for 50 hours and up to 50°C.
c. Repeat a and b with DC power applied to the DC injection coupler.
d. Remove the amplifier from the test setup and perform the flexibility test of 4.5.1. Repeat a at room temperature only.
e. Compare the test results with the phase delay variation requirement of 3.2.1.8.

4.5.12 Operating tests. Satisfactory operation after environmental testing shall be demonstrated by performing the gain tests of 4.5.2 at 10 MHz to 100 MHz only, the amplitude response test of 4.5.9 at 10 kHz only. After all tests, the sensitivity test of 4.5.8 shall be repeated at 10 kHz only.

4.5.13 Surface examination. Surface examination shall be performed in accordance with MIL-E-16400. Conformance to the requirements of 3.2.2.1, 3.3, 3.3.1, and 3.3.2, shall be verified during this examination. Radiographic inspection shall be conducted on the first article and on every fifteenth RF amplifier housing section to verify internal spacing and lay of reinforcing members and absence of flaws.

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4.5.14 Pressure resistance test. The RF amplifier, assembled with test cables as shown in FIGURE 13, shall be tested for watertight integrity as specified in a through c:

a. Remove the UG-1820/U connector from one of the approximately 30 ft test cables to permit placement in the pressure vessel.

b. Place the RF amplifier with test cables in a pressure vessel similar to that shown in FIGURE 14 with the free ends of the test cables in air at ambient pressure. Subject the RF amplifier to hydrostatic pressure specified in TABLE IV. The time to reach the required pressure shall be the minimum amount that the test equipment permits.

c. Upon removal from the pressure vessel the RF amplifier shall exhibit no evidence of physical damage or flooding and shall pass the operating tests of 4.5.13.

TABLE IV. Pressure resistance test requirements.

First article inspection		Production inspection	
Pressure (psig) 1/	Duration (minutes)	Pressure (psig)	Duration (minutes)
0	5	0	5
600	25	600	25
0	5	0	5
600	25	600	25
0	5	0	5
600	25	600	25
0	5	0	5
600	20 hours	600	25
0	5	0	5
600	25	600	25
0	5	0	5
600	25	600	25
0	5	0	5
		600	25
		0	5
		600	25
		0	5

1/ Pound-force per square inch gage

4.5.15 AN/BRA-24( ) durability test. The RF amplifier assembled with test cables shall be drawn 25 times in each direction through the Antenna Transfer Assembly, AN/BRA-24( ), as shown in FIGURE 15. The transfer assembly shall be filled with water at a hydrostatic pressure of 50 psig to 120 psig. The continuity of the assembly shall be monitored continuously during the test. Should the reading vary by more than ±10 percent from the initial value (nominal 400 ohms to 500 ohms), terminate the test and examine the assembly. Upon completion of passage through the AN/BRA-24( ), the RF amplifier shall exhibit no evidence of physical damage or flooding and shall pass the operating tests of 4.5.12.

4.5.16 Tensile test. The elongation and tensile strength of the RF amplifier shall be demonstrated by subjecting the RF amplifier to tensile loads along its axis in accordance with TABLE V. The test setup shall be similar to the one shown in FIGURE 16. The length of the housing shall be measured under load at the beginning (initial 10-pound (lb) load) and at the 500-lb tensile load during the fourth cycle. Upon completion of the tensile load cycles, the RF amplifier shall exhibit no evidence of physical damage and shall pass the electrical operating tests of 4.5.12. Elongation shall comply with the requirement of 3.2.3.5.

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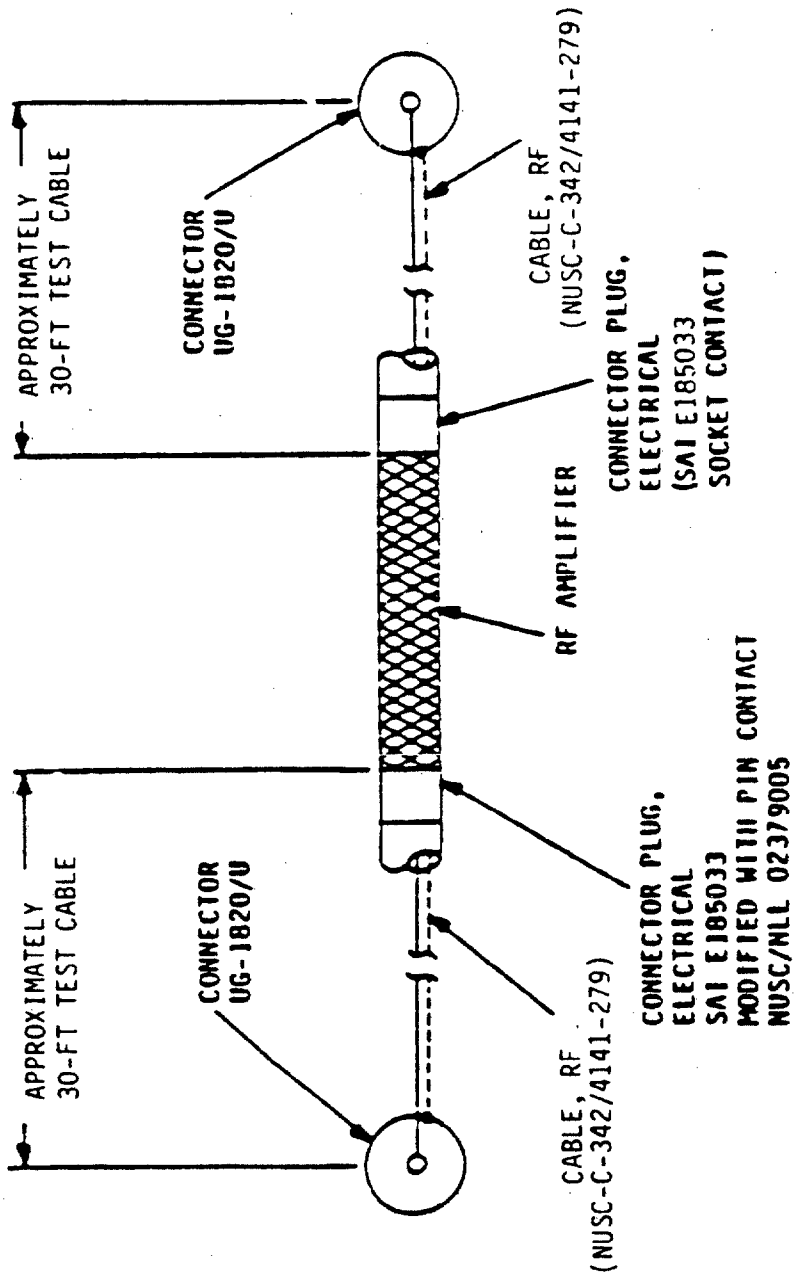


FIGURE 13. Test cable assembly.

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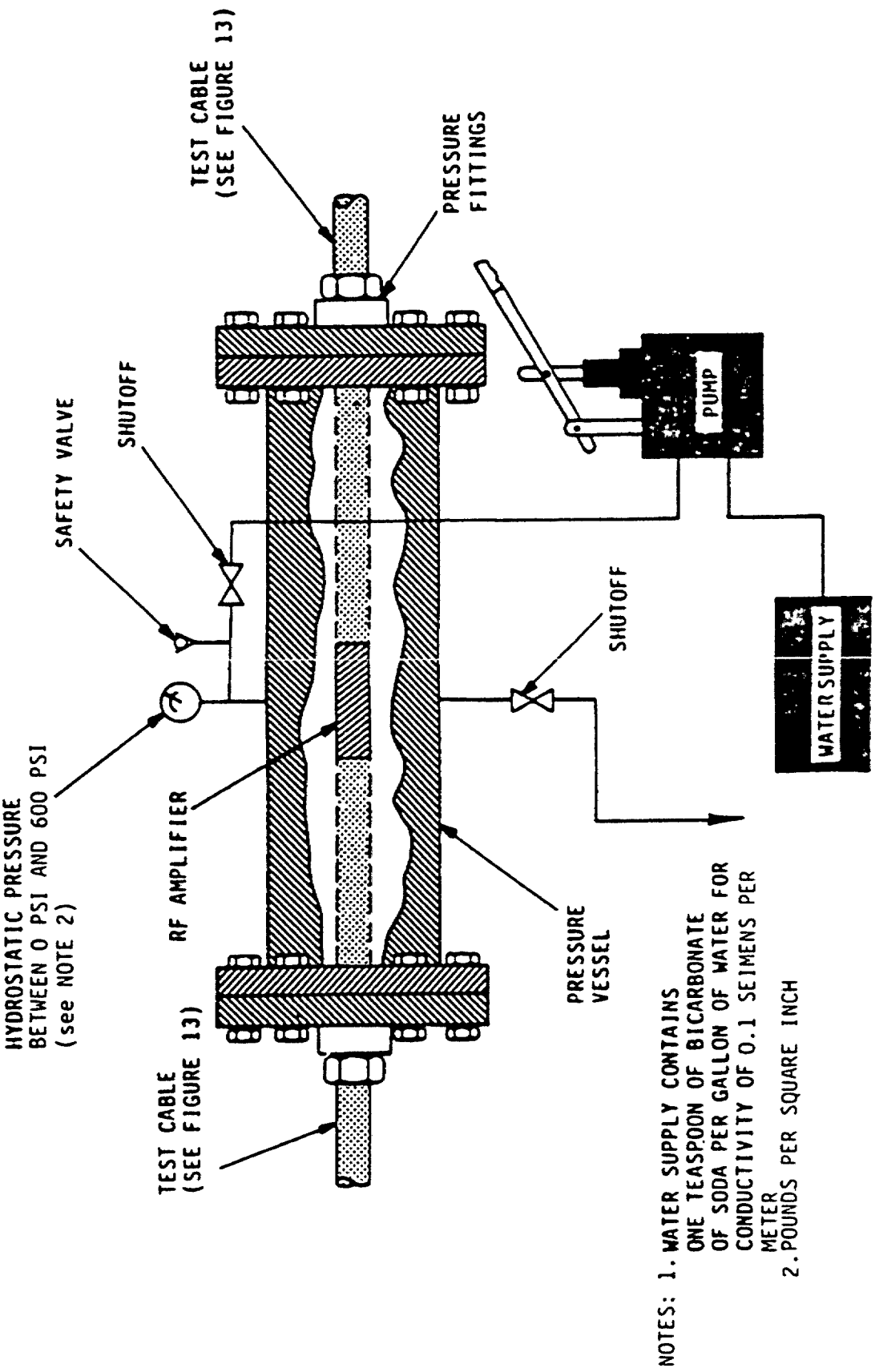


FIGURE 14. Pressure resistance test setup.

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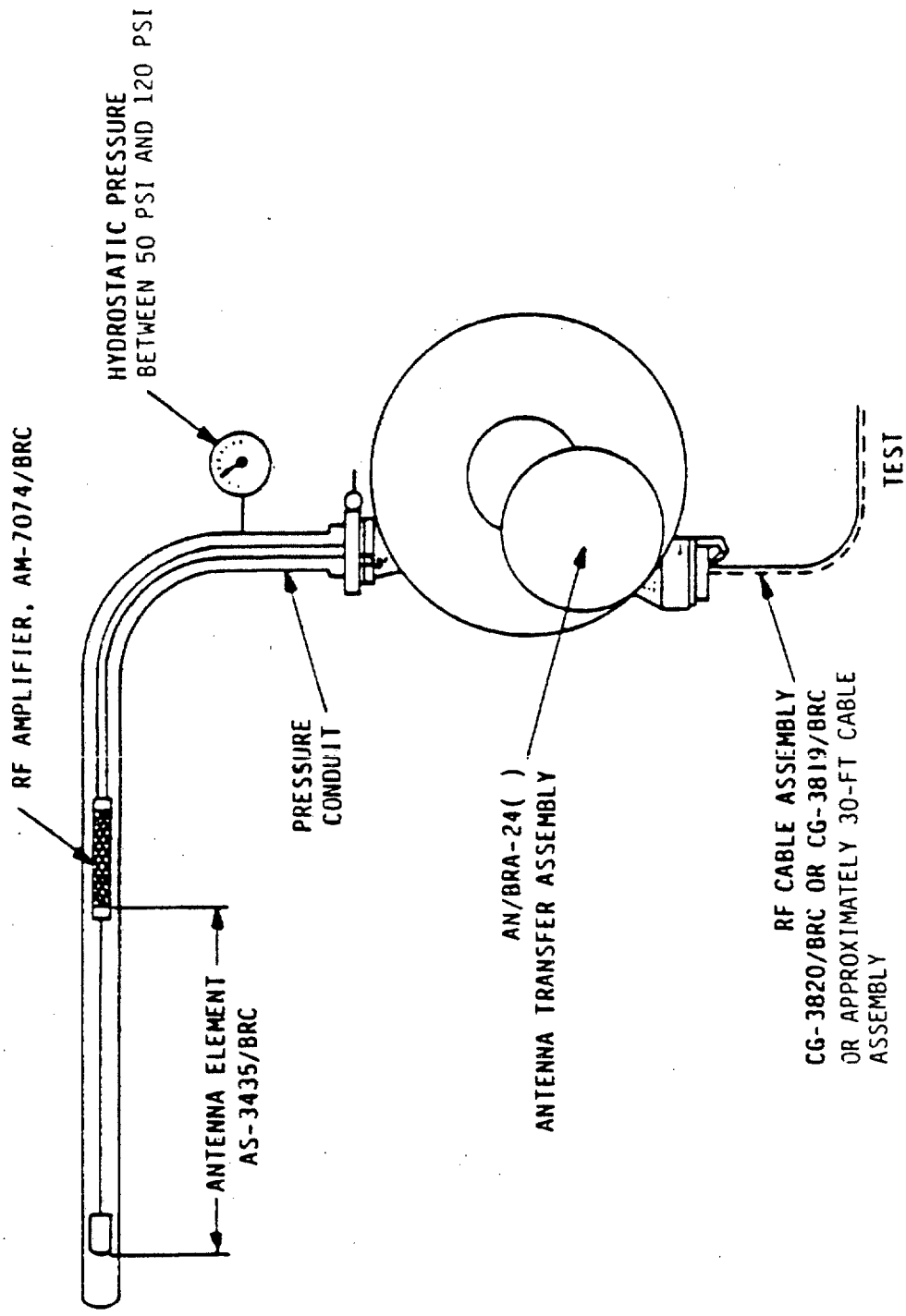


FIGURE 15. AN /BRA-24( ) durability test setup.

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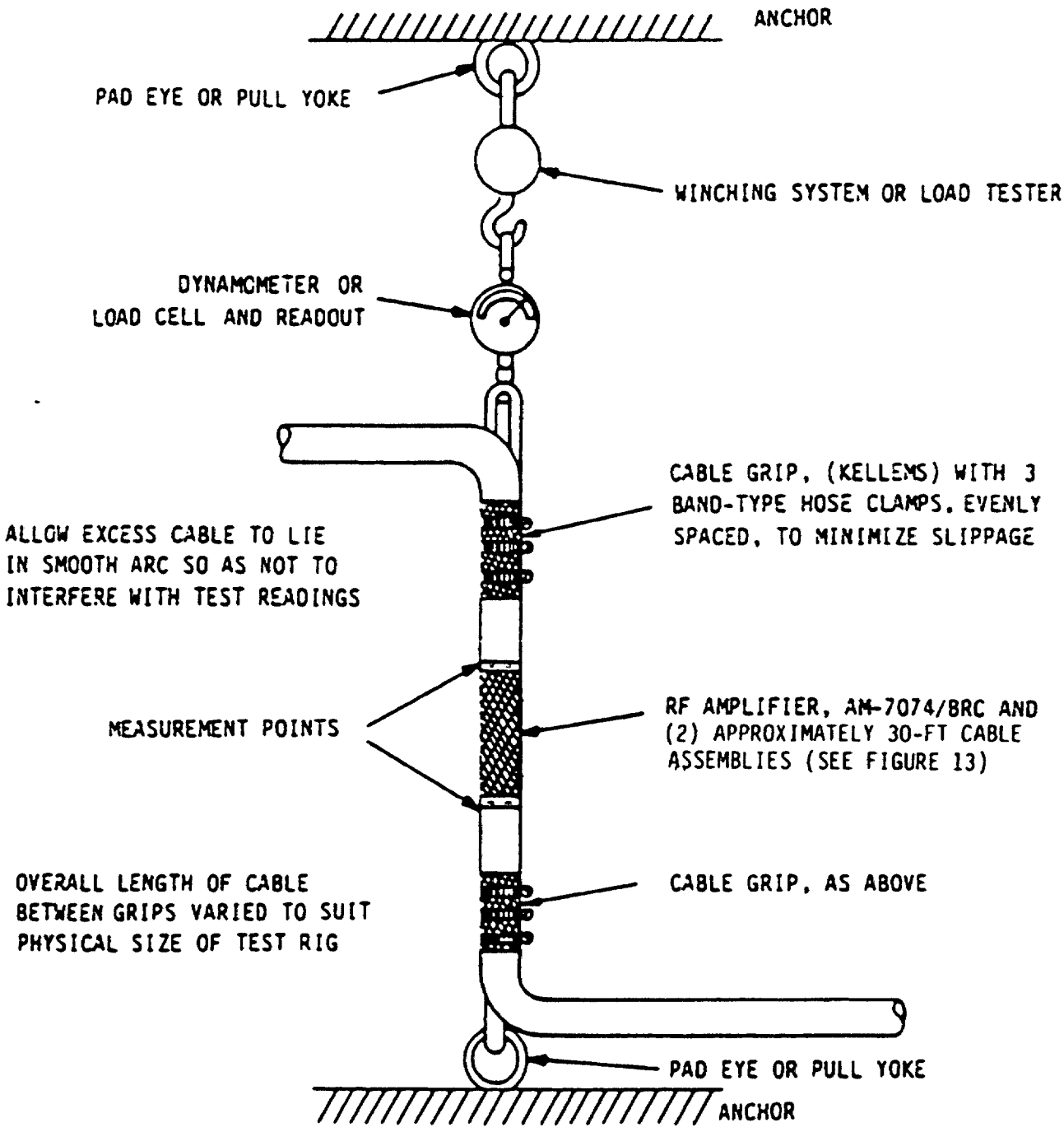


FIGURE 16. Tensile test setup.

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TABLE V. Tensile load test requirements.

Tensile load (lb)	Duration (minutes)
10	Time required to perform reference measurement
400	55
10	5
400	55
10	5
400	55
10	5
400	55
10	5

4.5.17 Diameter test. The RF amplifier shall be passed through a sizing block having a hole diameter of 0.670 in. to 0.671 in. to verify conformance to the requirements of 3.2.2.1.1.

4.5.18 Power supply interruption test. The RF amplifier, assembled with test adapters at each end, shall be tested for power supply interruption as specified in a through d:

a. Connect the RF amplifier in the test setup of FIGURE 4 with the spectrum analyzer replaced by a 50-ohm load. Apply DC power to the DC injection coupler.

b. Disconnect and reconnect the line between the RF amplifier and the DC injection coupler 100 times. The power supply current should drop to zero each time the line is disconnected and return to approximately 100 mA when it is reconnected.

c. Apply and remove a short circuit to the transmisison line between the RF amplifier and the DC injection coupler 100 times. The power supply should indicate 150 mA each time the short is applied and 100 mA when the short is removed.

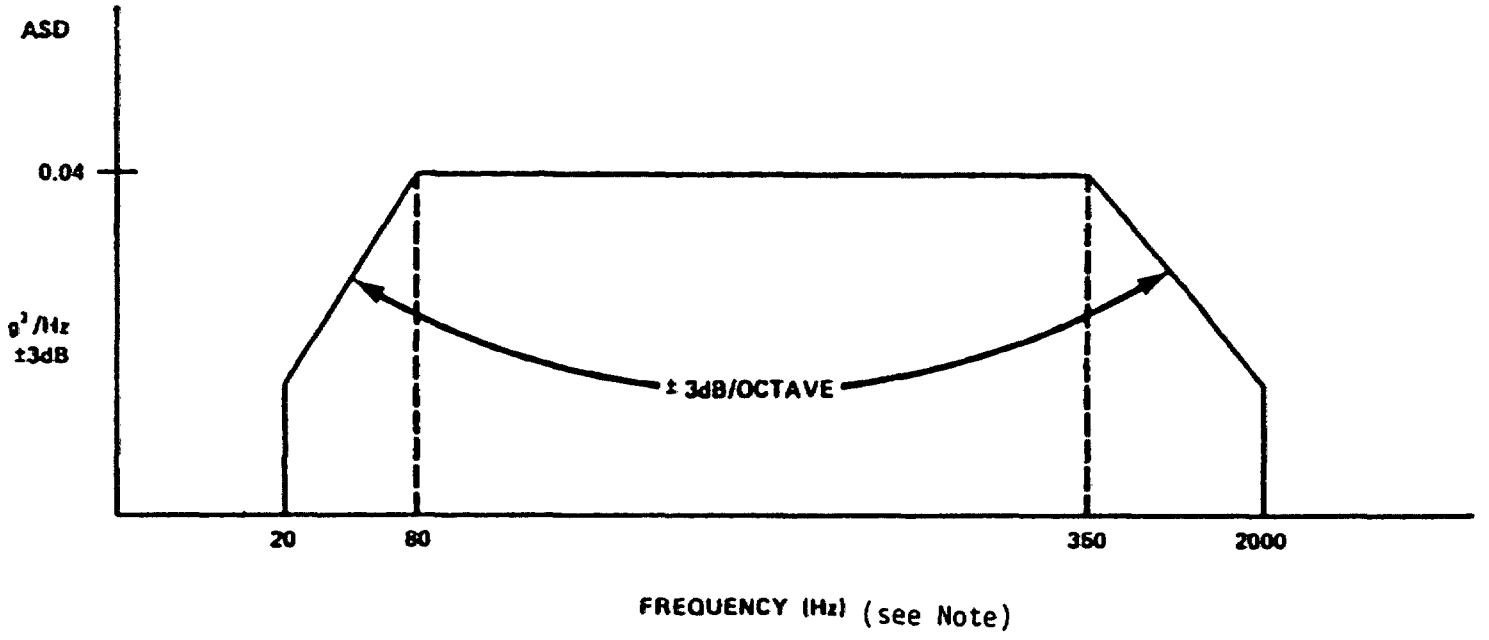
d. Subject the RF amplifier to the operating tests of 4.5.12 to confirm proper operation.

4.6 Workmanship. The RF amplifier shall be examined for workmanship to determine conformance to the surface examination requirements of 3.3.3 and the Workmanship paragraph of MIL-E-16400. Conformance to the requirements of 3.2.2, 3.3.1, and 3.3.2 shall be verified during this examination. Radiographic inspection shall be conducted on the first article and on every fifteenth RF amplifier housing section to verify correct internal spacing and lay of reinforcing members and the absence of flaws.

4.6.1 Workmanship screen. Vibration (see 4.6.1.1) and temperature cycling (see 4.6.1.2) tests shall be performed on each RF amplifier. Vibration shall be performed prior to temperature cycling. The vibration shall be performed on each RF amplifier.

4.6.1.1 Random vibration. The vibration shall be random, or subject to procuring activity approval, pseudo-random or complex waveform vibration, for an accumulated time of 10 minutes in the axis deemed most susceptible to vibration excitation. All items shall be hard-mounted (without shock isolators) and subjected to the vibration conditions of FIGURE 17. Input vibration levels shall be measured at the mounting points of the item under vibration. If variations are found at these points, the level used for control purposes shall be the average of the levels at the mounting points. Control equipment having a bandwidth no greater than 10 Hz for vibration frequencies up to 500 Hz, and 100 Hz for vibration frequencies above 500 Hz, shall be used for the control and analysis of the acceleration spectral density (ASD). The instantaneous acceleration peaks shall be limited to three times the rms acceleration level. The item shall be energized with 27 VDC supplied through the DC injection coupler (see FIGURE 3) during vibration and appropriate input signals applied to observe any abnormal conditions of the output functional characteristics. All failures occurring during screening shall be corrected and the vibration resumed.

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NOTE: Hertz

FIGURE 17. Random vibration curve.

4.6.1.2 Temperature cycling. Each RF amplifier shall be subjected to 10 cycles of the temperature curve shown in FIGURE 18. The temperature rate of change shall be not less than 5°C per minute. The RF amplifier shall be turned on and off at the indicated times. The RF amplifier shall be positioned for maximum exposure to the changing temperature. Where performance measurements are specified, a minimal functional operating test shall be performed. The dwell time shall be maintained until the largest electrical or electronic part in the RF amplifier reaches 80 percent of the chamber temperature. When failures occur, the RF amplifier shall be reworked and the cycling continued for a cumulative total of 10 cycles.

4.7 Inspection of preparation for delivery. Inspection shall be performed to ensure conformance with the requirements of Section 5.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Separate item. When delivered as a separate item, the RF amplifier shall be assembled in its housing with unmated ends capped, and wrapped with cushioning material (Acrotube or Armflex pipe insulation, or equal). A copy of the assembly instructions, NUSC/NLL DWG 02380006, shall be packaged with each RF amplifier in a plastic bag with the required screws and 50 percent spare screws. The RF amplifier shall be packaged in a box conforming to V_3 of PPP-B-636 with additional packaging to afford protection against physical damage during shipment, handling, and covered storage. The top and bottom flaps of the box shall be sealed by the application of 2-in. wide tape conforming to PPP-T-60, Type IV. The length of the box shall not exceed 12 in. length; 4 in. width; and 4 in. depth.

5.2 Part of assembly. When delivered with the associated antenna element, (AS-3435/8RC) or RF Cable Assembly, (CG-3819/8RC or CG-3820/8RC), or both, the RF amplifier shall be wrapped with cushioning material and coiled in the box specified for the antenna element or on the reel specified for the RF cable assembly. Unmated connectors shall be capped and required screws with 50 percent spare screws shall be supplied.

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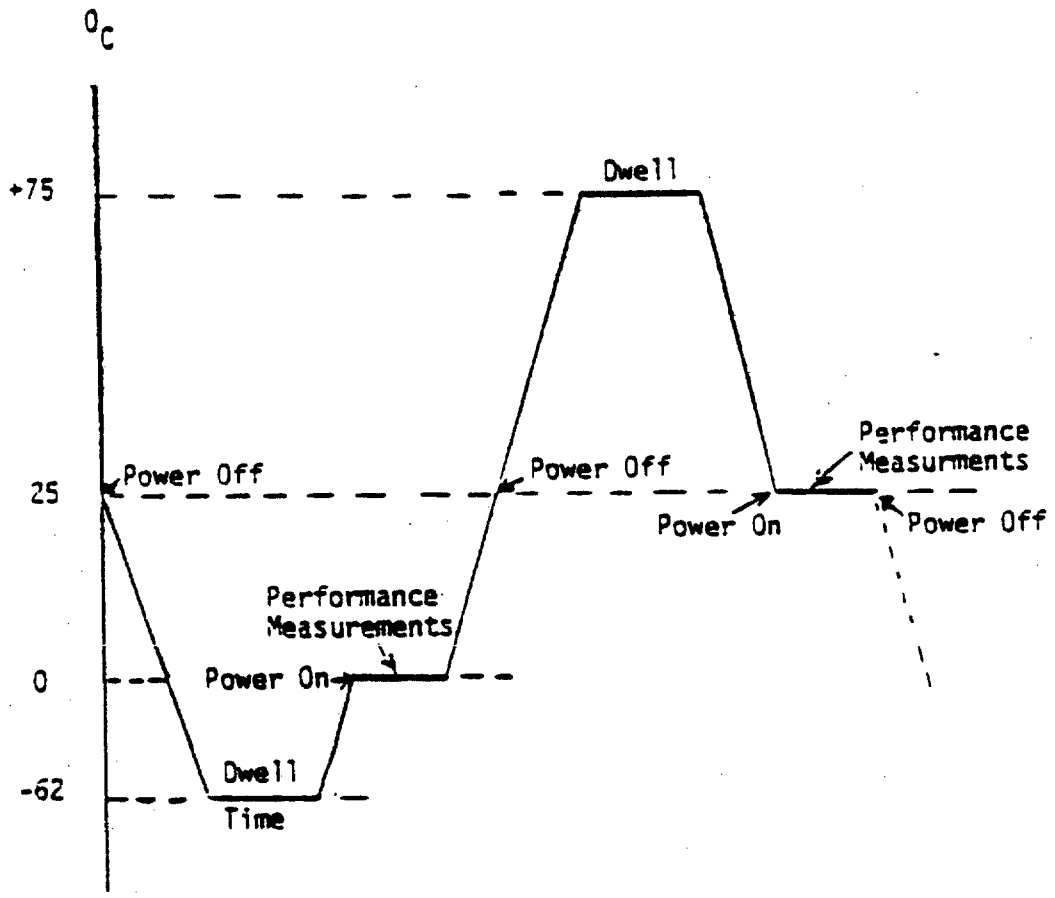


FIGURE 18. One cycle of temperature curve (not to scale).

6. NOTES

6.1 Intended use. The RF amplifier covered by this specification is intended for use by submarines as part of Buoyant Cable Antenna Assembly, AS-3434(V)/BRC.

6.2 Ordering data.

6.2.1 Procurement requirements. Procurement documents should specify:

a. Title, number, and date of this specification
b. Number of first article samples to be submitted if other than specified in 4.3
c. Procedures for equipment redesign in event of reliability failure (see 4.4.1.1)

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6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DID) (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List, incorporated into the contract. When the provisions of DoD Federal Acquisition Regulations Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs:

Paragraph	Data requirement	Applicable DID
4.3	First Article Inspection Procedure	DI-T-4901
4.3	First Article Inspection Report	DI-T-4902

(DIDs related to this specification, and identified in Section 6 will be approved and listed as such in DoD 5000.19L, Vol. II, AMSDL. Copies of DIDs required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 First article. When a first article inspection is required, the item(s) should be a first production item. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Supersession data. This specification includes the requirements of ELEX-A-526, dated 18 August 1982.

Preparing activity:
NAVY - EC

(Project No. 5985-N525-3(EC))

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-28897	Amplifier, Radio Frequency, AM-7074/BRC

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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